

Two-Factor RFID + Face Recognition Door

System Requirements & Software Requirements Specification

Document ID	DOC-01	Status	Implementation Baseline
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Engineering baseline: Build the smallest complete system that proves two-factor identity binding. Do not add cloud, multi-door, distributed, or enterprise infrastructure unless a requirement below changes.

1. Document Control

Field	Value
Document ID	DOC-01-SRS
Version	1.0
Status	Implementation Baseline
Date	September 29, 2026
Primary audience	Implementers, reviewer, AI coding agent, test operator
System type	Embedded + local computer-vision access-control prototype

2. System Objective

The system shall open one physical door only when a valid enrolled RFID card and the corresponding owner face are presented in the required sequence. A valid card starts a 10-second verification window. A matching face causes a grant; wrong face, no face, timeout, ambiguity, or required-component failure causes denial or fault-safe closure.

Canonical authorization rule

```
AuthorizationDecision =
  active_rfid
  AND verification_window_valid
  AND exactly_one_usable_face
  AND lbph_distance <= threshold
  AND recognized_person == card_owner
  AND stable_match_count >= required_consistent_matches
```

3. Scope and Non-Goals

In Scope	Explicitly Out of Scope
One door; one ESP32; one MFRC522; a small set of enrolled people; laptop webcam; local host	Multiple doors; high user counts; multi-site deployment
RFID detection + local identity lookup	Cryptographically strong smart-card authentication
OpenCV LBPH face recognition + enrollment	Liveness detection; anti-spoofing; depth/IR sensing
10-second verification window	Cloud recognition; remote authentication; mobile app
Servo/LED/buzzer control	Enterprise IAM, Kafka, Kubernetes, service mesh, HA clusters
SQLite/local file persistence and local logs	Distributed database or internet-dependent operation

Why this boundary matters: The project is small. Scalability is achieved through replaceable interfaces and explicit contracts, not by prematurely deploying infrastructure the MVP does not need.

4. Actors and Operational Roles

Actor	Purpose
Primary authorized user	Owns an enrolled RFID card and enrolled face.
Friend / test user	Provides a distinct enrolled face for negative-path testing.
Operator / implementer	Enrolls identities, configures the system, runs tests, and inspects logs.
ESP32 device	Reads RFID and drives physical indicators/servo.
Laptop host	Runs recognition, policy, persistence, and device communication.

5. Functional Requirements

ID	Requirement	Specification
FR-001	Startup and safe state	System initializes all required components and remains closed until the controller is healthy.
FR-002	RFID detection	ESP32 detects RFID card presentation using MFRC522 and reports the card

ID	Requirement	Specification
		UID to the host.
FR-003	RFID validation	Host accepts only active enrolled RFID UIDs and associates them with exactly one person.
FR-004	Verification window	A valid RFID event starts a 10-second face-verification window using a monotonic clock.
FR-005	Camera capture	Host acquires frames from the laptop webcam during the verification window.
FR-006	Face detection	Host determines whether zero, one, or multiple faces are visible in a frame.
FR-007	LBPH recognition	Host uses OpenCV LBPHFaceRecognizer for recognition of the detected face.
FR-008	Recognition threshold	LBPH distance is evaluated against a configured threshold; rejected results never grant access.
FR-009	Identity binding	Recognized face identity must equal the person bound to the presented RFID card.
FR-010	Stable-match gate	A grant requires N consecutive acceptable matches for the expected identity; default N=3.
FR-011	Wrong-face denial	A detected enrolled face that does not match the RFID owner causes immediate denial.
FR-012	No-face/timeout denial	No usable matching face before 10 seconds causes timeout and denial.
FR-013	Grant actuation	Only ACCESS_GRANTED causes the ESP32 to move the servo to the configured open position.
FR-014	Door hold and close	After grant, the door remains open for the configured hold interval, then returns to the closed position.
FR-015	Denied indication	Denial illuminates the red LED and produces a long buzz; servo remains closed.

ID	Requirement	Specification
FR-016	Granted indication	Grant illuminates the green LED and produces a short beep.
FR-017	Versioned protocol	Host/device communication uses versioned newline-delimited JSON messages with correlation IDs.
FR-018	Command acknowledgement	ESP32 acknowledges each valid actuator command so the host can distinguish accepted and failed execution.
FR-019	Fail closed	Unknown, stale, malformed, duplicate, or unactionable access commands do not open the door.
FR-020	Enrollment	Authorized operator can enroll a person by capturing face samples and associating one or more RFID cards.
FR-021	Access logging	Each authorization attempt records timestamp, card UID, expected identity, observed outcome, and reason.
FR-022	Model persistence	LBPH model metadata and trained model are saved locally and reloadable after restart.
FR-023	Health state	Host and ESP32 expose enough health/status information to prevent grants while required components are unavailable.
FR-024	Safe recovery	After restart or communication loss, the physical output returns or remains in the closed-safe state.
FR-025	Known spoofing limitation	Documentation shall explicitly state that printed-photo and presentation attacks are not reliably detected by this MVP.

6. Non-Functional Requirements

ID	Attribute	Specification
NFR-001	Responsiveness	The system starts face verification immediately after valid RFID acceptance and provides prompt local

ID	Attribute	Specification
		feedback.
NFR-002	Timing	Verification window is 10 seconds; timing uses monotonic time to avoid wall-clock changes.
NFR-003	Reliability	The system shall fail closed on missing camera, lost device link, malformed commands, or recognition uncertainty.
NFR-004	Maintainability	Recognition, authorization, persistence, communication, and hardware control are isolated behind interfaces.
NFR-005	Determinism	The authorization policy is explicit and testable; a face-recognition result never directly controls hardware.
NFR-006	Observability	Important state transitions, outcomes, failures, and correlation IDs are locally loggable.
NFR-007	Scalability seam	Repositories, transport, camera, recognizer, and device identity are replaceable without rewriting the core policy.
NFR-008	Local privacy	MVP operation requires no cloud service; biometric samples remain on the local host by default.

7. Security Requirements

ID	Requirement	Specification
SEC-001	Two-factor correlation	Grant requires both valid RFID possession and matching face identity for the same person.
SEC-002	Closed default	Power-up, restart, communication loss, and controller fault must not result in door opening.
SEC-003	Command safety	Only an authenticated-in-process authorization decision may produce ACCESS_GRANTED for the device adapter.
SEC-004	Replay/stale handling	Access commands carry

ID	Requirement	Specification
		session/command identifiers and expire with the access session.
SEC-005	Single decision point	One AccessController/AuthorizationPolicy owns the final decision; adapters cannot grant access independently.
SEC-006	RFID limitation	MVP treats card UID as an identifier, not as a cryptographically strong credential; cloning is out of scope.
SEC-007	Biometric limitation	LBPH is not liveness detection; printed photos and other presentation attacks may succeed.

Security posture: This is a prototype, not a production security product. Common RFID UIDs can be cloned, the USB/serial channel is trusted locally, and LBPH is not a liveness detector.

8. State Model

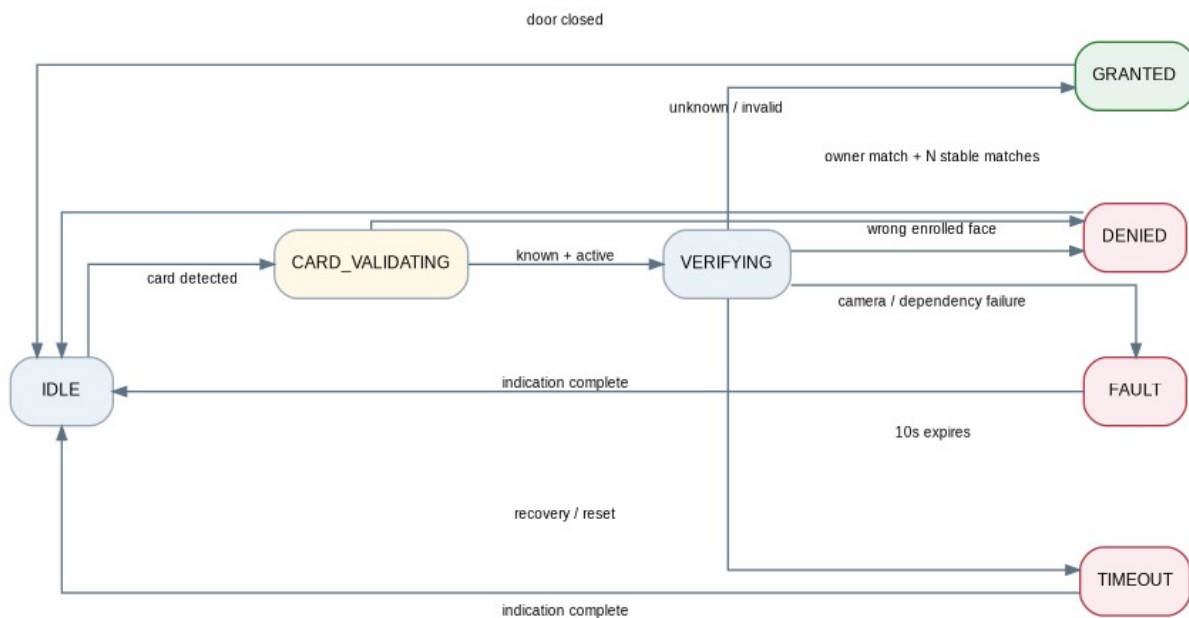


Figure 1. Canonical access-control state machine

State	Entry condition	Allowed outputs	Exit condition
IDLE	No active session	Servo closed; indicators off	RFID detected or fault
CARD_VALIDATING	RFID event received	No door open	Known active card -> VERIFYING; otherwise

State	Entry condition	Allowed outputs	Exit condition
			DENIED
VERIFYING	Known card + timer started	Camera capture only; no actuator grant yet	Owner match -> GRANTED; wrong face -> DENIED; 10s -> TIMEOUT; dependency failure -> FAULT
GRANTED	Authorization rule true	Open servo + green + short beep	Hold timer expires -> close -> IDLE
DENIED	Any explicit denial	Servo closed + red + long buzz	Indication complete -> IDLE
TIMEOUT	10s window expires	Servo closed + red + long buzz	Indication complete -> IDLE
FAULT	Required dependency unavailable	Servo closed; safe indication/log	Operator/system recovery -> IDLE

9. Access Flows

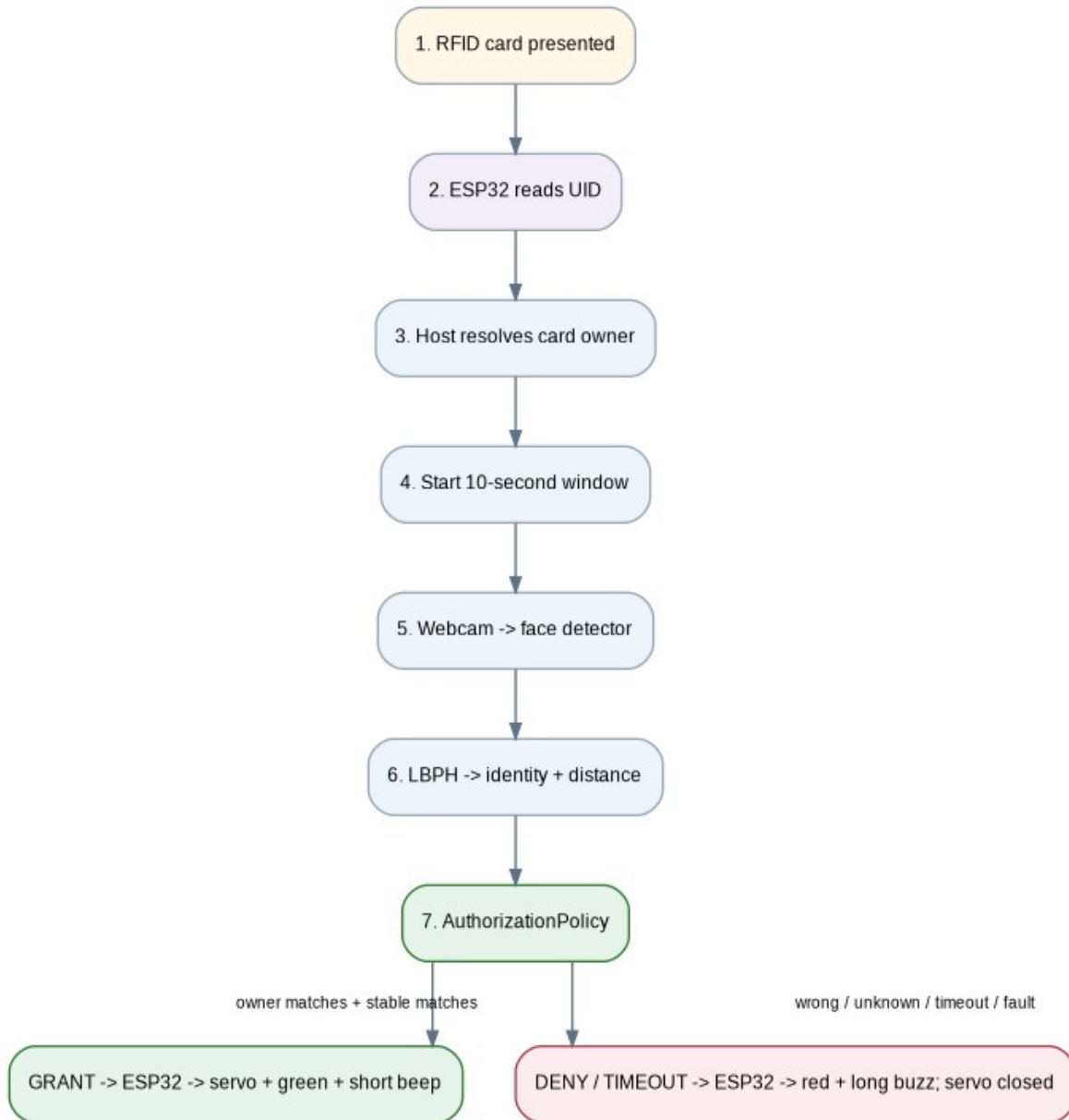


Figure 2. End-to-end authorization flow

10. Face Recognition Specification

The MVP uses OpenCV LBPHFaceRecognizer. The recognition stage is distinct from face detection: detection answers whether a face exists in a frame; LBPH answers which enrolled label is the closest match and exposes a distance/threshold mechanism. Recognition input is normalized grayscale face imagery.

```

frame -> detect face -> select usable face -> grayscale/normalize -> LBPH.predict -> (label, distance)
if distance > threshold: reject recognition result
if label != expected_face_label: deny
if label == expected_face_label for N consecutive samples: grant
  
```

The threshold is an empirical configuration value calibrated using the actual webcam, lighting, distance, and enrolled samples. The project shall not claim that one threshold is universally secure.

11. Enrollment Specification

1. Operator chooses a person identity.
2. Operator associates the RFID UID with that person.
3. Webcam captures multiple samples while a single face is visible.
4. Each sample is detected, cropped, converted to grayscale, and stored under the person identity.
5. LBPH model is trained/updated from the approved sample set.
6. Model metadata and identity-label mapping are persisted locally.
7. Operator performs a live verification test before considering the person active.

Enrollment rule: The system shall not treat a single captured photograph as a sufficient training set. Capture variation deliberately but keep the workflow simple for this MVP.

12. Protocol Requirements

Laptop and ESP32 communication shall use a versioned NDJSON protocol. Each message carries a type and, where relevant, event_id, session_id, and command_id so retries and stale commands can be identified.

ESP32 -> Host

```
{"v":1,"type":"rfid_detected","event_id":"evt-123","uid":"A1B2C3D4"}
```

Host -> ESP32

```
{"v":1,"type":"access_command","command_id":"cmd-456","session_id":"sess-123","decision":"grant","door_hold_ms":3000}
```

ESP32 -> Host

```
{"v":1,"type":"command_result","command_id":"cmd-456","result":"executed"}
```

ESP32 -> Host

```
{"v":1,"type":"heartbeat","device_id":"door-01","state":"CLOSED"}
```

13. Persistence Requirements

Entity	Required fields	Notes
Person	person_id, display_name, face_label, status, created_at	Identity used by both RFID and face bindings.
RFIDCard	rfid_uid, person_id, status, enrolled_at	MVP treats UID as identifier, not cryptographic proof.
FaceSample	sample_id, person_id, file_path, captured_at, quality/status	Local raw images used for training; operator controls enrollment.
SystemConfig	verification_window_s, door_hold_s, threshold, required_consistent_matches, camera_index, serial settings	Version/config validated at startup.
AccessEvent	event_id, timestamp, session_id,	Audit/debug record.

Entity	Required fields	Notes
	card_uid, expected_person, observed_person, result, reason	

14. Timing and Safety Rules

Rule	Specification
Verification window	10 seconds measured from valid RFID acceptance using monotonic host time.
Grant authority	Only AccessController/AuthorizationPolicy can produce a grant command.
Default physical state	Servo closed; no green success indication.
Failure state	Any unknown, stale, malformed, duplicated, or unacknowledged access command must not open the door.
Door hold	Default 3 seconds; configurable.
Stable matches	Default 3 consecutive acceptable LBPH matches.

15. Acceptance Criteria

ID	Scenario	Input	Expected Result
ACC-001	Correct identity	Enrolled card for Person A + Person A face within 10s	Green LED + short beep + servo opens
ACC-002	Wrong face	Person A card + Person B face	Red LED + long buzz + servo remains closed
ACC-003	Unknown card	Unknown card + any face	Denied without opening
ACC-004	No face	Person A card + no face for 10s	Timeout, red LED, long buzz, closed
ACC-005	Too slow	Person A card + correct face after 10s	Denied
ACC-006	Recognized wrong card	Person B card + Person A face	Denied
ACC-007	Multiple faces	Person A card + two visible faces	Denied; no grant
ACC-008	Camera failure	Valid card then camera	Denied/fault; door closed

ID	Scenario	Input	Expected Result
		unavailable	
ACC-009	Link failure	Valid card or grant while ESP32 link fails	No unsafe open; event logged
ACC-010	Photo spoof test	Person A card + printed Person A photo	May be accepted; limitation documented

16. Known Limitations and Security Disclosure

- A printed photograph may fool LBPH in this prototype. No liveness detection or presentation-attack detection is implemented.
- RFID card UIDs are identifiers, not strong cryptographic authentication. Cloning/replay resistance is out of scope for this MVP.
- The laptop is a trusted local control host. This is not a hardened physical security controller.
- Environmental changes in lighting, pose, camera angle, and enrollment quality can affect LBPH recognition.
- A servo mechanically attached to a door is only as secure as the physical enclosure and power design.

17. Traceability

Requirement group	Design artifact	Primary tests
FR-002..FR-004	Device Agent + AccessController + monotonic session timer	ACC-001..ACC-006
FR-005..FR-012	CameraPort + LBPH adapter + AuthorizationPolicy	ACC-001..ACC-007, ACC-010
FR-013..FR-016	Device command adapter + device firmware state	ACC-001..ACC-006, ACC-009
FR-017..FR-019	Protocol contract + command idempotency	ACC-008..ACC-009
FR-020..FR-022	EnrollmentService + repositories + model store	Enrollment integration tests
FR-023..FR-025	Health service + safety policy + documentation	Fault injection and photo test

18. References

OpenCV LBPHFaceRecognizer API: docs.opencv.org

NXP MFRC522 product page / lifecycle: [nxp.com](https://www.nxp.com)

NXP MFRC522 data sheet: [MFRC522 datasheet](#)

Espressif ESP32 UART documentation: docs.espressif.com